Wireshark Lab: HTTP v8.1
Questions and Answers

1. Is your browser running HTTP version 1.0, 1.1, or 2? What version of HTTP is the server running?

Answer: Both the browser and the server use HTTP/1.1.

2. What languages (if any) does your browser indicate that it can accept to the server?

Answer: English (en) and US English (en-US). Accept-Language: en-US,en;q=0.9.

3. What is the IP address of your computer? What is the IP address of the gaia.cs.umass.edu server?

Answer: Client: 10.0.0.44. Server: 128.119.245.12.

4. What is the status code returned from the server to your browser?

Answer: 200 OK.

5. When was the HTML file that you are retrieving last modified at the server?

Answer: Sat, 30 Jan 2021 06:59:02 GMT.

6. How many bytes of content are being returned to your browser?

Answer: 128 bytes.

7. By inspecting the raw data in the packet content window, do you see any headers within the data that are not displayed in the packet-listing window? If so, name one.

Answer: Yes. For example, Server, Last-Modified and Content-Type appear in the raw HTTP data but are not individually shown in the packet-listing window.

8. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE" line in the HTTP GET?

Answer: No. The first GET request does not contain an If-Modified-Since header.

9. Inspect the contents of the server response. Did the server explicitly return the contents of the file? How can you tell?

Answer: Yes. The response is 200 OK and contains the complete HTML file. The response body is 371 bytes long, as indicated by Content-Length: 371.

10. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an "IF-MODIFIED-SINCE:" line in the HTTP GET? If so, what information follows the "IF-MODIFIED-SINCE:" header?

Answer: Yes. The header is If-Modified-Since: Sat, 30 Jan 2021 06:59:02 GMT. It gives the last-modified date and time of the cached file.

11. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

Answer: 304 Not Modified. No file body is returned because the file has not changed; the browser uses its cached copy.

12. How many HTTP GET request messages did your browser send? Which packet number in the trace contains the GET message for the Bill of Rights?

Answer: One GET request, in packet 26.

13. Which packet number in the trace contains the status code and phrase associated with the response to the HTTP GET request?

Answer: Packet 32 in the reassembled HTTP display. The original status-line bytes are carried in packet 28.

14. What is the status code and phrase in the response?

Answer: 200 OK.

15. How many data-containing TCP segments were needed to carry the single HTTP response and the text of the Bill of Rights?

Answer: Four data-containing TCP segments: packets 28, 29, 31 and 32.

16. How many HTTP GET request messages did your browser send? To which Internet addresses were these GET requests sent?

Answer: Four visible HTTP GET requests: two to 128.119.245.12, one to 178.79.137.164 and one to 104.98.115.146. Three request the page and its images; the fourth is an OCSP certificate-status request.

17. Can you tell whether your browser downloaded the two images serially, or whether they were downloaded from the two web sites in parallel? Explain.

Answer: The two images are downloaded serially. The GET request for the second image is sent after the first image has been received.

18. What is the server's response (status code and phrase) in response to the initial HTTP GET message from your browser?

Answer: 401 Unauthorized.

19. When your browser sends the HTTP GET message for the second time, what new field is included in the HTTP GET message?

Answer: Authorization: Basic d2lyZXNoYXJrLXN0dWRlbnRzOm5ldHdvcms=

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